

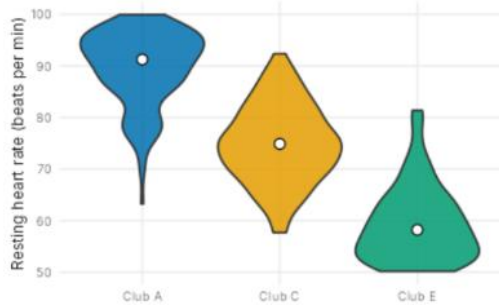
ACTIVITY 4 - QUESTION SHEET

DSA06 - Data Handling and Visualization | Activity: Identify Skewness using Violin Plots | Slot C

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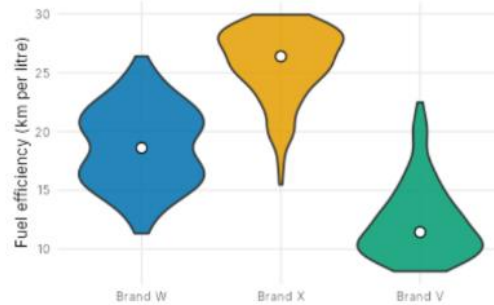
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Resting heart rate (beats per min)



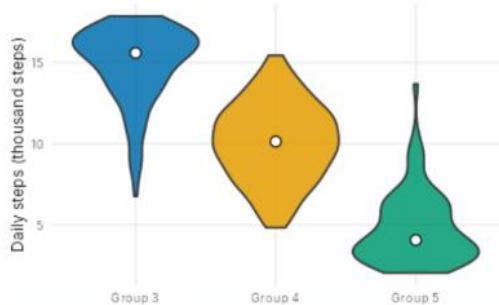
Violin	Skew (circle one)	Mean vs Median (circle)
Club A	Right / Left / Symm.	Mean > / < / ~ Median
Club C	Right / Left / Symm.	Mean > / < / ~ Median
Club E	Right / Left / Symm.	Mean > / < / ~ Median

Chart 2: Fuel efficiency (km per litre)



Violin	Skew (circle one)	Mean vs Median (circle)
Brand W	Right / Left / Symm.	Mean > / < / ~ Median
Brand X	Right / Left / Symm.	Mean > / < / ~ Median
Brand V	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Daily steps (thousand steps)



Violin	Skew (circle one)	Mean vs Median (circle)
Group 3	Right / Left / Symm.	Mean > / < / ~ Median
Group 4	Right / Left / Symm.	Mean > / < / ~ Median
Group 5	Right / Left / Symm.	Mean > / < / ~ Median

Chart 4: Monthly income (Rs thousand)



Violin	Skew (circle one)	Mean vs Median (circle)
Town	Right / Left / Symm.	Mean > / < / ~ Median
Metro	Right / Left / Symm.	Mean > / < / ~ Median
Suburb	Right / Left / Symm.	Mean > / < / ~ Median

Part B (2 marks)

B1. In Chart 3, violin "Group 3" is not symmetric. Towards which side is its tail, and which is the better measure of a typical daily steps here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

ASSESSMENT RECORD**Activity 4: Identify Skewness using Violin Plots | Slot C**

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Activity 4: Skewness in Violin Plots (R, ggplot2)
Name	BANDLAPALLI BHANUTEJA REDDY
Register Number	192325016
Date	08 OCTOBER 2026
Day	THURSDAY

Aim

To read twelve violin plots (four charts with three groups each) and decide for every violin whether the data is right-skewed, left-skewed or symmetric, to state how the mean compares with the median, and to explain each decision from the shape of the violin.

How I Read Skewness from a Violin Plot

A violin plot is a smooth outline of the data turned on its side. The width of the violin at any height tells how many values lie around that height, so a wide part means many values and a thin part means few. The white dot is the median. I use four simple clues:

Shape of the violin	Where the heavy part is	Tail	Skew	Mean vs Median
Heavy at the bottom, thin stretch upwards	Low values	Long, towards high values	Right (positive) skew	Mean > Median
Heavy at the top, thin stretch downwards	High values	Long, towards low values	Left (negative) skew	Mean < Median
Widest in the middle, both ends taper alike	Centre	Both sides similar	Symmetric	Mean $\approx$ Median

Why the mean moves. The median only cares about the middle position, but the mean uses every value. A few extreme values in a long tail pull the mean towards the tail, so the mean sits on the tail side of the median.

A common mistake to avoid. Skew is named after the side of the tail, not the side of the heavy part. A violin that is fat at the top with a tail going down is left-skewed, even though most values are high.

A note on the mean. A violin plot does not draw the mean, so I infer the mean-versus-median relation from the shape. The numbers I quote are approximate readings from the picture.

PART A: Skew and Mean vs Median for Every Violin (18 Marks)

Each chart below has a small answer table followed by a short explanation for every violin (1.5 marks per violin, 4.5 marks per chart).

Chart 1: Resting heart rate (beats per min)

Violin	Skew	Mean vs Median	Clue from the shape
Club A	Left skew	Mean < Median	Heavy at the top (about 85 to 100) with a thin tail down to about 63; median dot about 91.
Club C	Symmetric	Mean $\approx$ Median	Diamond shape, widest near 75, equal tapering to about 58 and 92.
Club E	Right skew	Mean > Median	Heavy at the bottom (about 50 to 60) with a thin tail up to about 82; median dot about 58.

Club A: Left skew, Mean < Median. Club A is fat at the top, between roughly 85 and 100 beats per minute, and then narrows into a thin tail that stretches down to about 63. The median dot (about 91) sits inside the heavy part near the top. The long tail points towards the lower values, so the few low heart rates pull the mean down and the mean ends up smaller than the median.

Club C: Symmetric, Mean $\approx$ Median. Club C looks like a diamond: it widens evenly towards the middle (around 75) and narrows at about the same rate towards 58 below and 92 above. The median dot sits close to the centre of the range and in the widest part. Both halves mirror each other, so neither side pulls the mean away and the mean and median are about equal.

Club E: Right skew, Mean > Median. Most of Club E is packed near the bottom, between about 50 and 60, and a thin tail reaches up to roughly 82. The median dot (about 58) lies low, in the heavy part. A few high heart rates in the upper tail drag the mean upwards, so the mean is greater than the median.

Result: In this chart the three clubs show all three patterns: Club A is left-skewed, Club C is symmetric and Club E is right-skewed.

Chart 2: Fuel efficiency (km per litre)

Violin	Skew	Mean vs Median	Clue from the shape
Brand W	Symmetric	Mean $\approx$ Median	Rounded top and bottom of similar size, range about 11 to 26.5; median about 18.5.
Brand X	Left skew	Mean $<$ Median	Heavy at the top (about 26 to 30) with a thin tail down to about 15.5; median about 26.3.
Brand V	Right skew	Mean $>$ Median	Heavy at the bottom (about 8 to 13) with a thin tail up to about 22.5; median about 11.5.

Brand W: Symmetric, Mean $\approx$ Median. Brand W has a small waist in the middle, but its upper and lower halves are rounded and of similar size, and the median (about 18.5) sits near the middle of the range from about 11 to 26.5. There is no long thin tail on either side, so I call it symmetric, with the mean close to the median.

Brand X: Left skew, Mean $<$ Median. Brand X is wide at the top, where the violin is cut off at about 30, and it thins out slowly downwards until about 15.5. The median dot (about 26.3) is in the heavy upper part. The long tail on the low side pulls the mean below the median.

Brand V: Right skew, Mean $>$ Median. Brand V is bulky near the bottom, between about 8 and 13 km per litre, and has a thin neck that rises to about 22.5. The median (about 11.5) is low. A few unusually efficient values in the upper tail raise the mean above the median.

Result: The three brands again show all three patterns: Brand W is symmetric, Brand X is left-skewed and Brand V is right-skewed.

Chart 3: Daily steps (thousand steps)

Violin	Skew	Mean vs Median	Clue from the shape
Group 3	Left skew	Mean $<$ Median	Heavy at the top (about 14 to 18) with a thin tail down to about 7; median about 15.5.
Group 4	Symmetric	Mean $\approx$ Median	Widest around 10, balanced tapering to about 5 and 15.5; median about 10.
Group 5	Right skew	Mean $>$ Median	Heavy at the bottom (about 3 to 6) with a thin tail up

			to about 13.5; median about 4.
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Group 3: Left skew, Mean < Median. Group 3 is wide near the top, around 14 to 18 thousand steps, with the top cut flat at about 18. A thin tail runs down to about 7 thousand. The median dot (about 15.5) is in the heavy part, and the few low-step people in the tail pull the mean below it.

Group 4: Symmetric, Mean $\approx$ Median. Group 4 has a smooth, balanced outline. It is widest around 10 thousand steps and narrows in a similar way towards about 5 below and about 15.5 above. The median dot is in the centre of the widest part, so the mean and median are about the same.

Group 5: Right skew, Mean > Median. Group 5 is packed near the bottom, between about 3 and 6 thousand steps, and a thin tail rises to roughly 13.5. The median (about 4) is low. The few very active people in the upper tail pull the mean above the median.

Result: Group 3 is left-skewed, Group 4 is symmetric and Group 5 is right-skewed, so the groups differ in shape as well as in level.

Chart 4: Monthly income (Rs thousand)

Violin	Skew	Mean vs Median	Clue from the shape
Town	Right skew	Mean > Median	Heavy at the bottom (about 15 to 40) with a thin tail up to about 102; median about 31.
Metro	Left skew	Mean < Median	Heavy at the top (about 120 to 150) with a thin tail down to about 67; median about 130.
Suburb	Symmetric	Mean $\approx$ Median	Widest around 70 to 80, balanced tapering to about 28 and 126; median about 76.

Town: Right skew, Mean > Median. In Town, most people earn between about 15 and 40 thousand, so the violin is fat at the bottom, and a thin tail stretches up to about 102 thousand. The median (about 31) is low. The few high earners in the tail pull the mean above the median, which is typical for income data.

Metro: Left skew, Mean < Median. Metro is wide near the top, around 120 to 150 thousand, where the violin is cut flat at about 150, and it narrows downwards to about 67. The median dot (about 130) is in the heavy part. The long low-side tail pulls the mean below the median.

Suburb: Symmetric, Mean $\approx$ Median. Suburb has an even, rounded outline that is widest around 70 to 80 thousand and tapers by similar amounts towards about 28 and 126. The median dot (about 76) sits close to the middle of the range, so the mean and median are about equal.

Result: Town is right-skewed, Metro is left-skewed and Suburb is symmetric, so one average income would not describe all three areas in the same way.

Summary Answer Key for All 12 Violins

Chart	Violin	Skew	Mean vs Median
Chart 1	Club A	Left skew	Mean < Median
Chart 1	Club C	Symmetric	Mean $\approx$ Median
Chart 1	Club E	Right skew	Mean > Median
Chart 2	Brand W	Symmetric	Mean $\approx$ Median
Chart 2	Brand X	Left skew	Mean < Median
Chart 2	Brand V	Right skew	Mean > Median
Chart 3	Group 3	Left skew	Mean < Median
Chart 3	Group 4	Symmetric	Mean $\approx$ Median
Chart 3	Group 5	Right skew	Mean > Median
Chart 4	Town	Right skew	Mean > Median
Chart 4	Metro	Left skew	Mean < Median
Chart 4	Suburb	Symmetric	Mean $\approx$ Median

Count: 4 right-skewed violins (Club E, Brand V, Group 5, Town), 4 left-skewed violins (Club A, Brand X, Group 3, Metro) and 4 symmetric violins (Club C, Brand W, Group 4, Suburb).

PART B (2 Marks)

B1: In Chart 3, violin "Group 3" is not symmetric. Towards which side is its tail, and which is the better measure of a typical daily steps value: mean or median? Give one reason. [1 Mark]

Answer: The tail of Group 3 is on the lower side (the left or downward side). The thin part of the violin stretches down towards about 7 thousand steps, so the group is left-skewed. The better measure of a typical value is the median.

Reason: The median is not pulled by the few extreme low values in the tail, so it stays at about 15.5 thousand steps, in the heavy part where most people are. The mean would be dragged downwards by those low values and would describe the group as less active than it really is.

B2: Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1 Mark]

```
ggplot(df, aes(x = group, y = value)) + geom_violin() + stat_summary(fun = median, geom = "point", shape = 21, fill = "white", size = 3)
```

How the line works: `aes(x = group, y = value)` puts the groups on the x-axis and the values on the y-axis. `geom_violin()` draws one violin for each group. `stat_summary(fun = median, geom = "point", ...)` calculates the median of each group and draws it as a white dot with an outline, which is exactly how the medians appear in the charts of this activity.

Another way: In older versions of ggplot2, `geom_violin(draw_quantiles = 0.5)` draws a line at the median inside each violin. I used `stat_summary` because it works in every version and gives the white dot.

Conclusion

The shape of a violin shows how the data is spread. A long thin tail upwards means right skew and a mean above the median, a long thin tail downwards means left skew and a mean below the median, and a balanced shape means the mean and median are close. When data is skewed, the median is the safer value to report as typical, because it is not pulled by a few extreme values.